

Safe, Green, and Efficient Synthesis of *m*-Dinitrobenzene via Two-Step Nitration in a Continuous-Flow Microreactor

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In industry, *m*-dinitrobenzene is mainly obtained by two-step nitration of benzene, but still faces the problem of being heterogeneous and highly exothermic. In this work, a two-step continuous-flow nitration was developed in a microreactor, combining benzene nitration and mononitrobenzene nitration in series. The waste acid after the mononitrobenzene nitration stage was utilized in the benzene mononitration stage, and the waste acid in the benzene mononitration stage was treated

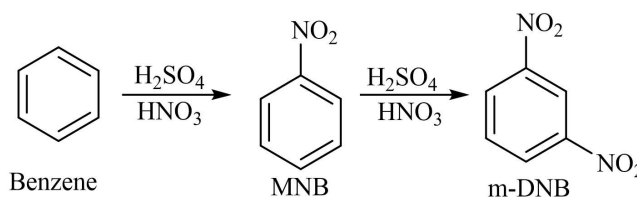
and concentrated for reuse. The whole process consumes almost no sulfuric acid. Under optimal conditions, the yield of dinitrobenzene was more than 99.0% and the selectivity of *m*-dinitrobenzene was more than 91% with a residence time of 4 min at 70 °C. In addition, the space-time yield of the continuous microreactor was 800 times higher than the batch reactor. It provides a safe and efficient method for the industrial production of *m*-dinitrobenzene.

1. Introduction

M-dinitrobenzene (*m*-DNB) is a very important basic industrial chemical, widely used in pesticide, photographic material, printing and dyeing industries. Its derivatives, including *m*-phenylenediamine, *m*-aminophenol and resorcinol, have many industrial applications. As we all know, the nitration of aromatic hydrocarbons is a reaction with a fast reaction rate and a violent exothermic reaction.^[1] And improper temperature control can lead to the decomposition of nitric acid and the generation of polynitro by-products.^[2] These factors pose a great challenge to the design of a highly safe and efficient reaction system.

Over the past two decades, the development of microreactor technology has demonstrated its unique advantages.^[3] The microreactor has the characteristics of intrinsic process safety, large specific surface area, excellent mass and heat transfer rate, accurate control over reaction parameters, high integration, and scalability. Due to the unique characteristics, the microreactor has a very good performance in the nitration process of aromatic hydrocarbons.^[4] Zhiquan Yu et al.^[5] developed a practical continuous nitration process for 2,5-difluoronitrobenzene via nitration of *p*-difluorobenzene with fuming nitric acid in 98% yield and further recycling of waste acid could be partially achieved by adjusting the concentration of sulfuric acid. Danilo et al.^[6] compared the performance of liquid/liquid biphasic benzaldehyde nitration in conventional batch reactor and continuous-flow microreactor and developed kinetic model under homogeneous conditions are validated

and extended for heterogeneous liquid-liquid systems. Dimitris et al.^[7] described the nitration of 2,4-dinitrotoluene (DNT) and its conversion to 2,4,6-trinitrotoluene (TNT) at a gram scale with the use of a fully automated flow chemistry system and the resulting process is safer, faster, and more efficient than previously reported TNT synthesis procedures. Zhenghui et al.^[8] investigated the continuous-flow nitration of trifluoromethoxybenzene in a microchannel reactor and presented a scale-up strategy combining microreactor with distributed packed tubular reactor was presented. In industrial, *m*-DNB is mainly obtained by two-step nitration of benzene (Scheme 1). The nitration of *m*-DNB in microreactors has rarely been reported, especially synthesis process at kilogram-scale. Many research groups have investigated benzene nitration reactions in microreactors, but all of them are at laboratory scale.^[9] Some academics have also studied straightforward one-step dinitration of aromatic in continuous-flow reactors.^[10] This process usually requires about 40%–50% excess nitric acid, and the reaction heat can be twice as much as mononitration. This will lead to the raising of production cost, is not suitable to produce *m*-DNB. Moreover, *m*-DNB has a low melting point and is prone to precipitate solids during the reaction (< 65 °C), which affects the stability of the system. In summary, the development and design of the nitration of *m*-DNB in a microreactor has the most promising potential for industrial production.



Scheme 1. Two-step nitration of benzene to prepare *m*-DNB.

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Supporting information for this article is available on the WWW under <https://doi.org/10.1002/slct.202204997>

In this work, a process for the preparation of m-DNB by two-step nitration in a continuous flow microreactor was developed. Firstly, the conditions of benzene mononitration stage and mononitrobenzene(MNB) nitrification stage were explored respectively, including reaction temperature, the molar ratio of nitric acid to MNB or benzene and mass fraction of sulfuric acid in the mixed acid, and the total flow rate. Then, benzene nitration and mononitrobenzene nitration were connected in series (Figure 1). In the MNB nitrification stage, MNB and mixed acid were produced in the microreactor to produce m-DNB and waste acid, then the reaction solution flowed into the liquid-liquid separator. The waste acid and the organic phase were easier to separate due to their different densities. The waste acid was reused in the benzene nitrification stage after the addition of nitric acid. The organic phase became the product after post-treatment. In the benzene nitrification stage, benzene and mixed acid react in the microreactor to produce nitrobenzene and waste acid. The waste acid from the benzene nitrification stage was reused after simple treatment and concentration. This method improved production efficiency and made the reaction milder and more controllable.

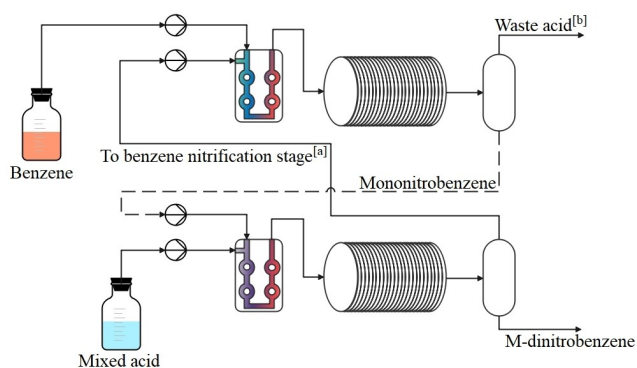


Figure 1. Process for producing m-DNB by two-step continuous-flow at kilogram-scale; [a] the waste acid in the mononitrobenzene nitration stage can be used as a mixed acid for benzene nitration after adding nitric acid; [b] the waste acid (mainly sulfuric acid) after nitration of benzene can be recycled after being treated and concentrated.

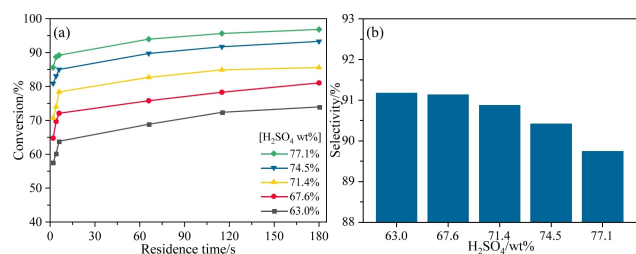


Figure 2. Effects of the mass fraction of sulfuric acid in mixed acid on (a) conversion of MNB, (b) selectivity of m-DNB; 65 °C, molar ratio(HNO₃/MNB) = 1.0, total flow rate = 30.0 mL/min.

2. Results and Discussion

2.1. Optimization of key operating conditions for MNB nitration stage

2.1.1. Effects of the mass fraction of sulfuric acid in mixed acid on MNB nitration stage

By varying the mass fraction of sulfuric acid (63%–77.1%) in the mixed acid, the effects of sulfuric acid strength were investigated at 65 °C, molar ratio (HNO₃/MNB) = 1.0 and the total flow rate was 30.0 mL/min. Figure 2(a) indicates the conversion of MNB increased with the increasing sulfuric acid mass fraction and residence time. Due to nitro can passivate the benzene ring, resulting in a slower nitration reaction of MNB, more sulfuric acid is required to increase the nitration capacity of the mixed acid.^[11] On the one hand, sulfuric acid as a catalyst to promote the nitric acid's protonation and promote the production of NO₂⁺.^[12] On the other hand, increasing the amount of sulfuric acid as water-absorbing agent reduced the dilution effect of the reaction produced water during the reaction process.^[13] Figure 2(b) shows that the m-DNB selectivity decreased slightly with increasing sulfuric acid mass fraction, dropping from an average of 91.5% to about 89.75%. The analysis of reaction products indicated that the side reactions mainly included ortho-, para-position nitration and nitrophenols. In general, the higher sulfuric acid strength, the easier to formation of trinitrobenzene by-product.^[14] The reactants were precisely controlled by the microreactor with excellent mass transfer and mixing conditions so that no trinitrobenzene were found.

2.1.2. Effects of the molar ratio of HNO₃/MNB on MNB nitration stage

By varying the molar ratio of HNO₃ to MNB (1.0–1.25), the effect of HNO₃ strength was investigated at 65 °C, 71.4% sulfuric acid mass fraction, and the total flow rate was 30.0 mL/min. As shown in Figure 3(a), the conversion of MNB increased with the increase of molar ratio and residence time. Theoretically, the molar ratio of HNO₃ to MNB of 1.0 is sufficient. However, the generation of water in the reaction will weaken the nitrification ability of the nitrifying agent, which will eventually make it difficult to completely convert the feedstock. Therefore, increasing the molar amount of nitric acid can increase the

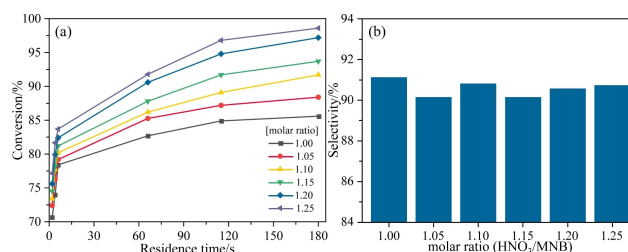


Figure 3. Effects of molar ratio of HNO₃/MNB on (a) conversion of MNB, (b) selectivity of m-DNB; 65 °C, 71.4% sulfuric acid mass fraction, total flow rate = 30.0 mL/min.

reaction rate of MNB. The rate of NO_2^+ generation is linked to the concentration of nitric acid. Sufficient nitric acid can accelerate the reaction rate and shorten the reaction time. However, the excessive use of nitric acid will increase the post-treatment difficulty and production cost. As shown in Figure 3(b), the selectivity of m-DNB was not significantly affected by the molar ratio. This is mainly due to the positive kinetic effect promoting both the primary and secondary reactions as the molar ratio increases. The adjacent- and para- position nitration reactions of MNB are generally considered as parallel side reactions, which react in the same order as the main reaction.^[8]

2.1.3. Effects of temperature on MNB nitration stage

For the heterogeneous nitration reaction, temperature significantly affects the mass transfer between the two phases and the reaction kinetics. The effects of temperature on the MNB nitration stage were quantified by varying the temperature (65–82 °C) in the reactor with 71.4% sulfuric acid mass fraction, molar ratio (HNO_3/MNB) = 1.20, and the total flow rate was 30.0 mL/min. Figure 4(a) shows that the conversion of MNB increased by increasing either the residence time or the reaction temperature. The former may be attributed to the longer contact time of the two reactant phases and the longer reaction time. According to the Arrhenius equation, the reaction rate coefficient increases with increasing temperature. In addition, since the molecular diffusion coefficient is proportional to the reaction temperature, the increasing temperature can enhance the mass transfer across the multiphase interface, which can further increase the apparent reaction rate. As shown in Figure 4(b), the selectivity of the m-DNB decreases with increasing temperature, suggesting that side reactions are accelerated at higher temperatures and increase later in the reaction. On the one hand, high temperatures lead to nitric acid decomposition, and the pressure in the reaction must be controlled to reduce nitric acid decomposition. On the other hand, the lower reaction temperatures can also lead to the precipitation of MNB, which can easily clog the microreactor. The product was able to dissolve when the temperature exceeded 65 °C and the system ran smoothly. In general, the higher the temperature, the higher the risk of reaction. The

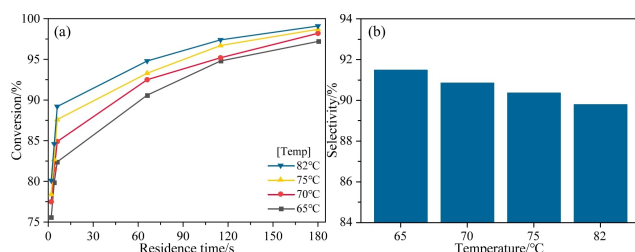


Figure 4. Effects of reaction temperature on (a) conversion of MNB, (b) selectivity of m-DNB; 71.4% sulfuric acid mass fraction, molar ratio (HNO_3/MNB) = 1.2, total flow rate = 30.0 mL/min.

microreactor has intrinsic safety characteristics and is safer than traditional tank reactors under the same conditions.

2.1.4. Effects of varying total flow rate on MNB nitration stage

This possible effect of total flow rate was investigated by varying the total flow rate in the range of 10–50 mL/min. Accordingly, the length of the delay loop tube was changed to maintain a constant residence time (180 s) at 70 °C, 71.4% sulfuric acid mass fraction, molar ratio (HNO_3/MNB) = 1.2. Figure 5 shows the effects of the total flow rate and residence time. In the low total flow rate range (< 30 mL/min), the conversion increased from 91.7% to 98.9% and the selectivity also increased from 84.9% to 90.6% as the total flow rate increased. When the flow rate was in the high flow range (> 30 mL/min), the conversion decreased slightly, and the selectivity remained largely unchanged. The variation in reactant conversion was probably a result of the effect of the total flow rate on the flow dynamic behaviour of droplet-based microfluidics, such as the geometry of the dispersed droplets, their possible coalescence along the microchannel and the two-phase interface property (Figure S4, SI).^[15] Two-phase nitrification occurs mainly at the interface of the two phases, which leads to the thermal effect of nitrification being highly concentrated. Typically, smaller dispersed droplets can be formed at high total flow rates, increasing the surface area of the two phases, and resulting in better mass transfer performance and higher reaction rates.^[16] The separation index X_s can be used to characterize the mixing effect of a microreactor (Figure 6). The higher the flow rate, the lower X_s of the microreactor, indicating that its mixing effect is better.^[17] However, as the flow rate was further increased, the coalescence of the dispersed droplets occurred resulting in the decrease of the two-phase interfacial area and reduced mass transfer performance, despite the possible chaotic flow of the continuous phase at high flow rates.^[18] When the total flow rate

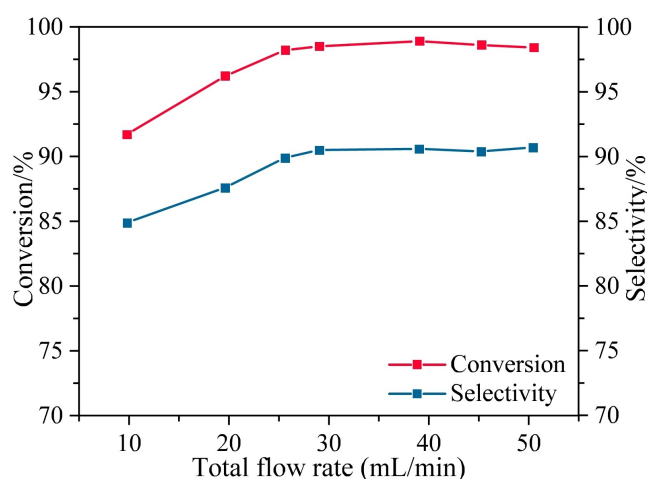


Figure 5. Effects of the total flow rate on conversion of MNB and selectivity of m-DNB; 70 °C, 71.4% sulfuric acid mass fraction, molar ratio (HNO_3/MNB) = 1.2, residence time = 180 s.

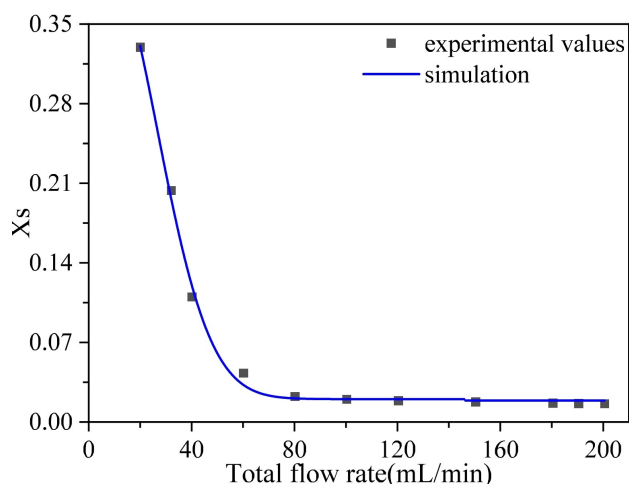


Figure 6. Variation of X_s with the total flow rate of the microreactor.

was low (<30 mL/min), the mass and heat transfer in the microreactor was not optimal, resulting in more by-products. When the mass transfer reached better conditions, the nitration reaction remained approximately constant under these operating conditions. Overall, the microreactor's precise control of the reactant flow rate provides for a flexible operating space that facilitates optimization of operating conditions and process scale-up.

2.2. Optimization of key operating conditions for benzene mononitration stage

Industrially, m-DNB is produced from benzene by a two-step nitration process. First, benzene was nitrated to produce MNB, and then MNB was further nitrated to produce m-DNB. It is necessary to investigate the nitration of benzene to produce MNB in the continuous microreactor. In general, nitration of benzene is easy to perform, and the by-product is mainly a small amount of dinitrobenzene. This section investigates the nitrification of benzene in a plate microreactor suitable for industrial production. It examines the conditions of temperature, sulfuric acid mass fraction, molar ratio (HNO_3/MNB), and total flow rate to provide a reference for the industrial production of benzene nitrification.

2.2.1. Effects of temperature on benzene mononitration stage

Figure 7 indicates the effect of the temperature on benzene conversion and selectivity of MNB. The temperature significantly affected the nitrification reaction, and the benzene conversion increased with increasing residence time at different temperatures. Moreover, the reaction temperature in the range of 40 and 70 °C, the selectivity of MNB decreased from 99.3% to 98.0%. In industrial production, the temperature of benzene nitration is typically 70–75 °C to convert benzene to MNB as much as possible. As the batch reactor cannot precisely control the temperature, more by-products are produced. Due

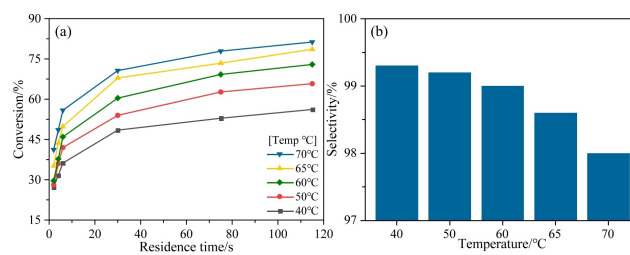


Figure 7. Effects of temperature on (a) conversion of benzene, (b) on selectivity of MNB; 71.4% sulfuric acid mass fraction, molar ratio (HNO_3/MNB) = 1.0, total flow rate = 30.0 mL/min.

to the excellent heat and mass transfer capabilities of the microreactor, the temperature is lower and more uniformly distributed, reducing the occurrence of by-products.

2.2.2. Effects of mixed acid on benzene mononitration stage

Figure 8 illustrates the effect of the sulfuric acid mass fraction in mixed acid on the benzene conversion and selectivity of MNB. The higher sulfuric acid strength in the mixed acid, the higher of conversion and impurity, and the lower product's selectivity. When the sulfuric acid mass fraction was 36.4%, only half of the benzene was converted, but with the highest selectivity. When the sulfuric acid mass fraction reached 51.2%, the benzene conversion reached 95.6% and the selectivity of MNB was 98.4%. Similar to the MNB nitration stage, the content of by-products in the reaction increases slowly with increasing sulfuric acid mass fraction, and the selectivity decreases slowly.

Figure 9 shows the effect of molar ratio (HNO_3/MNB) on the conversion of benzene and the selectivity of MNB. The conversion of benzene was positively correlated with the molar ratio, the benzene conversion reached 99% when the molar ratio was above 1.2. And the selectivity of benzene nitration was less affected by the molar ratio and was around 99%. As the molar ratio increases, the concentration of NO_2^+ ions in the mixed acid continues to increase. Although a high molar ratio could promote the conversion of benzene, excess nitric acid causes waste and complicates the post-treatment.

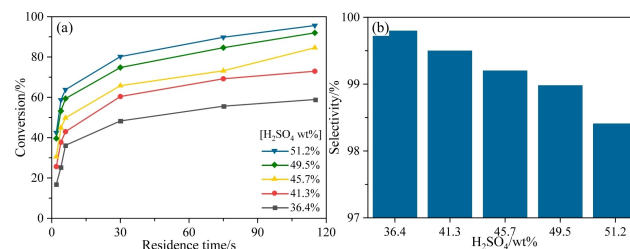


Figure 8. Effects of sulfuric acid mass fraction in mixed acid on (a) conversion of benzene, (b) on selectivity of MNB; 60 °C, molar ratio (HNO_3/MNB) = 1.0, total flow rate = 30 mL/min.

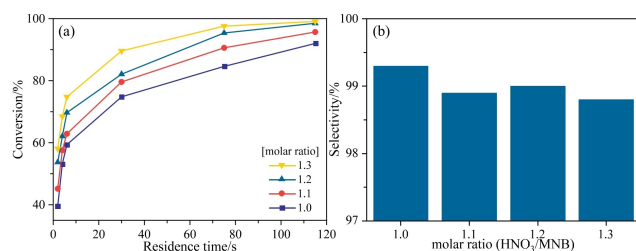


Figure 9. Effects of molar ratio of HNO_3 to MNB on (a) conversion of benzene, (b) selectivity of MNB; 60°C , 49.5% sulfuric acid mass fraction, total flow rate = 30.0 mL/min.

2.2.3. Effects of varying total flow rate on benzene mononitration stage

Figure 10 represents the effect of the total flow rate in mixed acid on the conversion of benzene and the selectivity of MNB. When the flow rate was less than 30 mL/min, the conversion rate and selectivity increased as the flow rate increased. When the flow rate continued to increase, the conversion rate and selectivity remained unchanged. Higher flow rates improve the

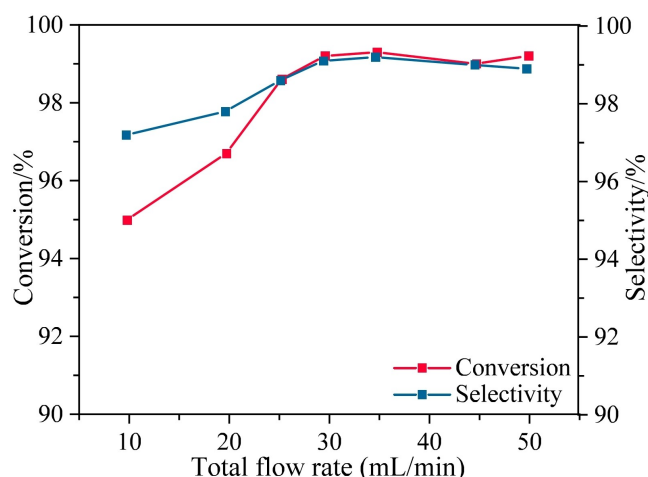


Figure 10. Effects of the total flow rate in mixed acid on conversion of benzene and selectivity of MNB; 60°C , 49.5% sulfuric acid mass fraction, molar ratio (HNO_3/MNB) = 1.2, residence time = 115 s.

mixing effect of the microreactor and enhance mass transfer. When the flow rate was further increased, the mixing effect of the reactants was no longer the main condition affecting the reaction, so the conversion rate remained mainly the same.

2.3. Preparation of m-DNB by two-step continuous nitration at kilogram-scale

Based on the experimental results of the previous work, a process for the kilogram-scale production of m-DNB by two-step continuous nitration was developed. First, the nitration of MNB was carried out with a total flow rate of 45 mL/min (residence time was 230 s), and then dinitrobenzene and mixed acid were separated at the end of the reaction. Then, the waste acid after the MNB nitration stage was added to nitric acid to make molar ratio of HNO_3 to MNB was 1.05, and then used as a mixed acid for the benzene nitration stage. Finally, the total flow rate in the nitration of benzene was 45 mL/min (residence time was 130 s), and the MNB and mixed acid were separated at the end of the reaction. As shown in Table 1, under optimal conditions, the conversion of MNB was 99.5% and the selectivity of m-DNB was 90.88% in the MNB nitration stage; the conversion of benzene was 99.5% and the selectivity of MNB was above 99% in the benzene mononitration stage. The nitration conditions of benzene are relatively mild. Although the waste acid after the MNB nitration stage contains a certain amount of water, which weakens the nitration ability, it is still suitable for the nitration reaction of benzene. In the benzene mononitration stage, nitric acid can be almost completely consumed, which makes subsequent operations more convenient. More importantly, the waste acid after benzene nitration (almost exclusively sulfuric acid) can be recycled after treatment and concentrated. In the post-treatment stage, the organic phase from the MNB nitration stage was sent to the product tank after deacidification, neutralization, and washing, and sulfuric acid can also be recovered (Figure S5, SI). In the sulfuric acid concentration stage, about 85% or more of the sulfuric acid can be recovered after reduced pressure distillation. Therefore, almost no sulfuric acid is consumed in the whole process, and the excess of nitric acid is only 1–2%. This makes the process very economical. Although the MNB synthesized by the actual reaction was not used in the MNB nitration stage, due to the high conversion rate and high selectivity in the benzene mononitration stage,

Table 1. Results of different reaction conditions.

Table 1. Results of different reaction conditions.											
item	MNB nitration stage				benzene mononitration stage						
	T [°C]	sulfuric acid mass fraction[%]	molar ratio [HNO ₃ /MNB]	Dinitrobenzene yield [%]	M-DNB selectivity [%]	nitric acid in waste acid [equiv]	added nitric acid [equiv]	T [°C]	conversion [%]	MNB selectivity [%]	nitric acid in waste acid [equiv]
1	65	67.8	1.15	95.8	91.8	0.19	0.86	65	98.6	99.5	0.03
2	65	70.6	1.15	98.3	91.6	0.14	0.86	65	99.0	99.0	0.01
3	65	67.8	1.20	97.1	90.9	0.23	0.82	70	98.7	99.4	0.04
4	65	72.3	1.20	99.7	90.8	0.20	0.85	70	99.5	99.2	0.02
5	70	71.4	1.15	97.8	91.5	0.15	0.90	65	99.2	99.3	0.01
7	70	71.4	1.20	99.5	91.0	0.19	0.86	70	99.5	99.6	0.02

the difference between the two was minimal and does not affect the actual reaction. Furthermore, the by-products are mainly para- and ortho-dinitrobenzene, which still have economic value after distillation and purification.

2.4. Comparison of experimental results between batch reactor and microreactor

The continuous-flow process was compared with the batch reaction process, and the results are shown in Table 2. The conversion of benzene in the microreactor was similar to the batch reactor, above 99.0%. Since the reaction conditions could be precisely controlled and the occurrence of side reactions was reduced, the yield of dinitrobenzene was more than 99.0% and the selectivity of m-DNB was 90.88%. During the nitration reactions, the formation of self-reactive nitrophenols is difficult to avoid.^[19] 2,4,6-trinitroresorcinol, a major by-product in the dinitrobenzene production process, is self-reactive substance and has led to tragic explosion incidents in China.^[20] Notably, the amount of nitrophenol is very small and must be concentrated 50 times before it can be detected. The continuous-flow process reduced the reaction time from 180 min to 4 min and lowered the temperatures. The mixed acid is reused for the nitration of benzene after the reaction reducing the material and energy consumption of the waste acid recovery process. The instantaneous volume in the microreactor was only 200 mL, so the safety of the reaction can be well guaranteed. In addition, the space-time yield of the continuous microreactor was $803.1 \text{ mol L}^{-1} \text{ h}^{-1}$ (800 times than batch reactor). The production rate in the microreactor was approximately 1.62 kg/h, and industrial-scale production can be realized by the numbering-up principle. Compared with the batch reactor, microreactor has the advantages in terms of safety, high production efficiency, and easier production scale-up than batch reactions.

3. Conclusion

In this work, the preparation of m-DNB by two-step nitration in a continuous-flow microreactor was carried out. The main operating parameters were explored to quantify their effect on

the reaction results, including reaction temperature, sulfuric acid mass fraction, molar ratio (HNO_3/MNB), and total flow rate. Furthermore, the production of m-DNB by two-step nitration of benzene was investigated on kilogram-scale, and the waste acid concentrate and reuse were realized. In the MNB nitration stage, the conversion of MNB reached 99.5% and the selectivity of m-DNB was 90.88%; in the benzene mononitration stage, the conversion of benzene was 99.5% and the selectivity of MNB was above 99%. Compared to the batch reaction, the residence time was shortened from 180 to 4 minutes, and the instantaneous volume was only 200 mL. Moreover, the productivity of m-DNB reached 1.62 kg/h, and the space-time yield was $803.1 \text{ mol L}^{-1} \text{ h}^{-1}$ (800 times than batch reactor). The process realizes green chemical synthesis by recycling sulfuric acid and consuming only nitric acid in the reaction process. The excellent yield, high throughput, safety and controllability of the proposed process indicated its significant industrial application potential.

Experimental Section

Materials

Chemicals used in this work mainly included: fuming nitric acid (> 98) sulfuric acid (> 98), sodium bicarbonate (> 99), dichloromethane (≥ 99.9), benzene (> 99), Mononitrobenzene (MNB, > 99). The deionized water was made in-house. All chemicals and reagents were purchased from Sinopharm Chemical Reagent Co., Ltd, and were used as received without any further purification.

Experimental setup and procedures

Structure of microreactor

As shown in Figure 11, the microreactor consists of a micromixer and the delay loop tube. The micromixer consists of two serpentine heat exchange channels sandwiching the reactant mixing channel, with optimized split-and-recombine structures. The primary body material of the micromixer was fabricated by Hartz alloy (HC276). The overall shape of the micromixer is a thin rectangular plate with a characteristic size of $85 \text{ mm} \times 65 \text{ mm}$, containing multiple

Table 2. The results of batch reactor and microreactor.					
item	T [°C]	Reaction time [min]	Dinitro- benzene yield [%]	M-DNB selectivity [%]	Space-time yield [$\text{mol L}^{-1} \text{ h}^{-1}$]
Microreactor ^[a]	70	4	≥ 99.0	90.88	803.1
batch reac- tor ^[b]	75- 85	180	98.0	85.00	1.07

[a] MNB nitration stage: 70 °C, 71.4% sulfuric acid mass fraction, molar ratio (HNO_3/MNB) = 1.20; benzene mononitration stage: 70 °C, add 0.86 equivalent of nitric acid to make molar ratio (HNO_3/MNB) was 1.05.
[b] MNB nitration stage: 85 °C, 73.5% sulfuric acid mass fraction, molar ratio (HNO_3/MNB) = 1.08, benzene mononitration stage: 75 °C, 45.2% sulfuric acid mass fraction, molar ratio (HNO_3/MNB) = 1.05.

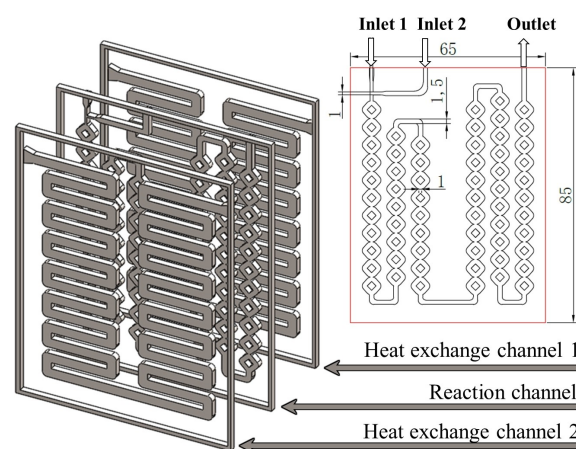


Figure 11. Dimensional structure diagram of the microreactor.

rhomboid split mixing channels in series, the internal channel width is 1.5 mm, and the contact channel of the two fluids is 1 mm. The two different flows are recombined and reshaped in the rhombic sub-channels to achieve efficient mixing.^[21] A delay loop tube (316 stainless steel, inner diameter of 2 mm, length 12.7–76.4 m) was connected to the outlet of the micromixer to ensure sufficient residence time. Multiple sampling points were established for the micromixer and delay loop tube to obtain samples at different residence times.

Experimental procedure for nitration of benzene/MNB

The same microreactor and experimental approach were used for the benzene mononitration and MNB nitration stages. As shown in Figure 12, two high-precision piston pumps (Sanotac ISP1020) were used to transport the organic phase and the mixed acid through the respective inlet tube into the microreactor and then into the delay loop tube. Both the inlet and delay loop tubes were placed in a thermostatic water bath (Julabo GmH CORIO CD-200F) at preset reaction temperature. The residence time was controlled by the length of the reaction channel or the flow rate. The mixed acid solution was configured according to the amounts of nitric acid and sulfuric acid. The amount of sulfuric acid was determined as the mass fraction of sulfuric acid to the mixed acid. The ratio of nitric acid to MNB/benzene was controlled by the volumetric flow rate ratio of the two phases. When the system was stable, samples were taken at each sampling port of the microreactor. After sampling, the samples were quickly added to the saturated NaHCO₃ solution after being taken out. After the acid was completely neutralized, the organic phase was separated and then washed with deionized water, and then the sample was extracted by adding dichloromethane. Three samples were taken at each sampling point, and the results were averaged. At the end of the reaction, the product and the waste acid are separated in a micro liquid-liquid separator connected to the outlet of the time delay tube.

Experimental procedure for the preparation of m-DNB at kilogram-scale

As shown in Figure 1, m-DNB was produced by two-step nitration at the kilogram-scale in a microreactor. Equal amounts of MNB and benzene were used, and nitration experiments were performed separately. First, nitration of MNB was carried out according to the experimental conditions with a total flow rate of 45 mL/min (residence time was 230 s). Then the content of nitric acid in the waste acid was measured, and the nitric acid was supplemented so that the molar ratio of HNO₃ to MNB was 1.05. Finally, the mixed acid was used to nitrify benzene with a total flow rate of 45 mL/min (residence time was 130 s). At the end of the reaction, the waste acid is separated from the product using a

liquid-liquid separator. In real industrial production, the transfer of MNB and the replenishment of sulfuric acid with waste acid can be easily achieved. Moreover, the concentration and recovery of sulphuric acid is a relatively mature process.

Analytical methods

Prepared product solutions were diluted with dichloromethane for the determination of the products distribution by gas chromatograph with a flame ionization detector (Agilent 7890 A, column: DB-5 30 m × 0.25 mm × 0.25 μm, and the film thickness: 1.0 μm) with nitrogen as the carrier gas. The temperature of the column was initiated at 60 °C, and then increased to 200 °C with the speed of 10 °C per minute, whereas those of the injector and the detector were kept at 280 °C and 300 °C, respectively. The carrier gas was fed with the speed of 6.5 mL/min, while there were 400 mL/min for air and 40 mL/min for hydrogen for running the flame ionization detector. The injecting volume of the sample was 1.0 μL. The area normalization method was used to determine the concentration of the components quantitatively. The maximum error of the measurements did not exceed 3%. The respective retention time of components are benzene (1.785 min), MNB (7.342 min), p-dinitrobenzene (p-DNB, 11.981 min), m-dinitrobenzene (m-DNB, 12.263 min), o-dinitrobenzene (o-DNB, 12.491 min).

Supporting Information Summary

The supporting information includes detailed batch experimental procedures and compound characterization data (GC), droplet status during the reaction, sulfuric acid concentration and product post-treatment.

Acknowledgements

The authors are grateful for financial supports from the National Natural Science Foundation of China (U20A20143).

Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available in the supplementary material of this article.

Keywords: Microreactors · Continuous flow · Nitration · Kilogram-scale · m-dinitrobenzene

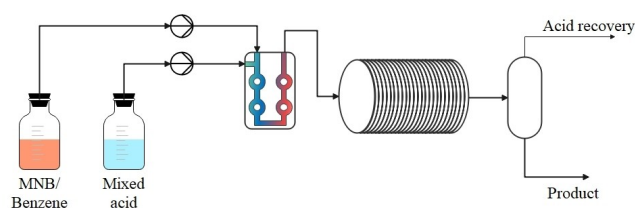


Figure 12. Process of continuous nitration of MNB/benzene.

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Submitted: December 26, 2022

Accepted: March 27, 2023